

SPACECRAFT DYNAMICS CAPSTONE PROJECT

Attitude Dynamics and Control of a Nano-Satellite Orbiting Mars

COURSE REPORT

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Abstract

This project presents the development and validation of a complete attitude dynamics and control framework for a nano-satellite operating in low Mars orbit. The mission requires autonomous transitions between multiple operational modes, including Sun-pointing for power generation, nadir-pointing for scientific observations, and GMO-pointing for communication with a geostationary Mars orbiter.

Mission-specific reference frames were derived analytically from the orbital geometry and implemented in MATLAB to generate the desired spacecraft attitude commands. The spacecraft rotational dynamics were modeled using rigid-body equations of motion, while attitude representation and tracking were performed using Modified Rodrigues Parameters (MRPs). A proportional-derivative (PD) attitude-control law was designed using a worst-case inertia approach to achieve a desired closed-loop decay time constant of 120 seconds.

Numerical simulations were conducted to evaluate spacecraft performance during individual pointing modes as well as throughout a complete autonomous mission scenario. The results demonstrate successful attitude stabilization, effective detumbling, and accurate tracking of both inertially fixed and rotating reference frames. Following operational-mode transitions, transient tracking errors were rapidly damped and converged in a manner consistent with the controller design objectives. The complete mission simulation further verified the ability of the spacecraft to autonomously switch between operational modes while maintaining stable closed-loop behavior.

To provide independent validation of the numerical results, the mission scenario was reconstructed in Systems Tool Kit (STK) using the simulated attitude histories. The STK visualization confirmed the spacecraft orientation and pointing behavior during all mission phases and showed strong agreement with the MATLAB simulations. Overall, the project demonstrates the successful integration of spacecraft attitude dynamics, autonomous control, mission-level decision logic, and three-dimensional visualization for a multi-mode Mars-orbiting nano-satellite.

Chapter 1

Introduction

This project considers a small science satellite orbiting Mars in a Low Mars Orbit (LMO), tasked with collecting scientific data from the Martian surface and relaying the acquired information to a larger mother spacecraft orbiting in a Geostationary Mars Orbit (GMO). In addition to science and communication operations, the spacecraft must periodically orient its solar panels toward the Sun to maintain sufficient onboard power.

Accordingly, the spacecraft mission is divided into three operational modes:

- 1) **Science Mode:** pointing the sensor toward the Martian surface,
- 2) **Power Mode:** orienting the solar panels toward the Sun,
- 3) **Communication Mode:** pointing the communication antenna toward the GMO spacecraft.

Both the LMO and GMO spacecraft are assumed to follow known circular orbits whose orientations are described using Hill reference frames generated through a 3-1-3 Euler-angle sequence.

The LMO spacecraft is equipped with:

- A science sensor aligned with the $+b_1$ body axis,
- Solar panels whose normal vector aligns with the b_3 axis,
- A communication antenna aligned with the $-b_1$ axis.

The objective of this project is to design and validate an attitude determination and control framework capable of autonomously switching between the required operational pointing modes.

Chapter 2

Mission Scenario Definitions

2.1 Mission Orbit Overview

The LMO and GMO spacecraft are both assumed to follow circular orbits around Mars. The orbital motion is modeled using classical two-body dynamics, where Mars is treated as the central gravitational body.

The physical and orbital parameters used throughout the simulations are summarized in Table 2.1.

Table 2.1: Simulation and Spacecraft Parameters

Parameter	Value	Unit
Mars radius R_{Mars}	3396.19	km
Mars gravitational parameter μ_{Mars}	4.28283×10^{13}	m^3/s^2
LMO altitude	400	km
LMO orbital radius r_{LMO}	3796.19	km
GMO orbital radius r_{GMO}	20424.2	km
LMO orbital angular rate $\dot{\theta}_{LMO}$	0.000884797	rad/s
GMO orbital angular rate $\dot{\theta}_{GMO}$	0.0000709003	rad/s

The orbital angular rates are computed using the circular-orbit relation:

$$\dot{\theta} = \sqrt{\frac{\mu_{Mars}}{r^3}} \quad (2.1)$$

The initial orbit orientation angles of both spacecraft are summarized in Table 2.2.

Table 2.2: Initial Orbit Frame Orientation Angles

Spacecraft	Ω (deg)	i (deg)	$\theta(t_0)$ (deg)
LMO	20	30	60
GMO	0	0	250

Figure 2.1 illustrates the definition of a circular orbit using relative elements along with inertial and Hill frames' base vectors. Note that the orbital elements (Ω, i, θ) can be used as (3-1-3) Euler Angles to obtain Hill frame $\mathcal{H}$ from inertial frame $\mathcal{N}$.

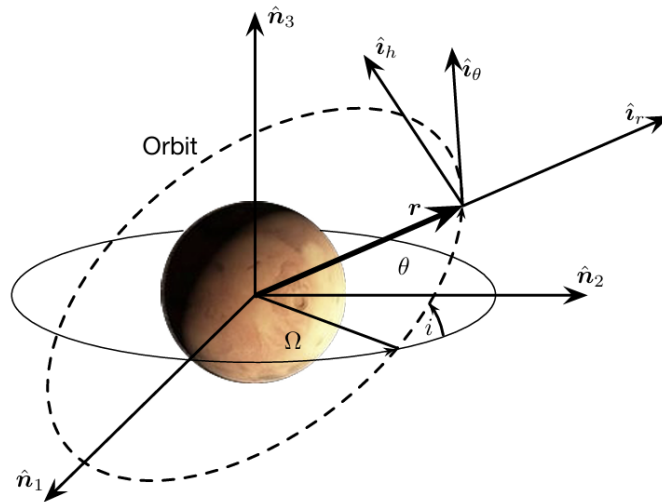


Figure 2.1: Illustration of circular orbital elements alongside inertial frame $\mathcal{N} : \{\hat{n}_1, \hat{n}_2, \hat{n}_3\}$ and Hill frame $\mathcal{H} : \{\hat{i}_r, \hat{i}_\theta, \hat{i}_h\}$ base vectors

2.2 Mission Pointing Scenarios

A body frame as shown in Figure 2.2 is mounted on the spacecraft, assumed to be the principal body frame.

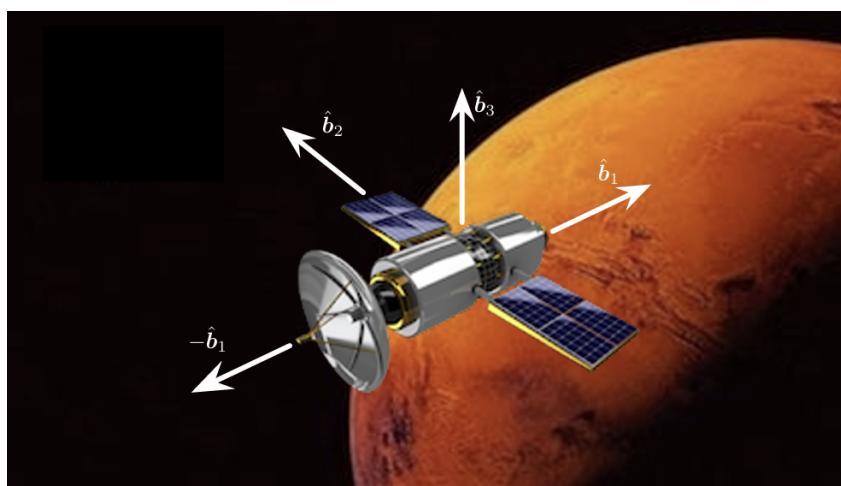


Figure 2.2: Illustration of LMO spacecraft's body frame $\mathcal{B} : \{\hat{b}_1, \hat{b}_2, \hat{b}_3\}$

Three operational spacecraft pointing modes are defined according to mission requirements.

The Sun is assumed to be located at an effectively infinite distance and aligned with the inertial n_2 axis. Therefore, during Sun-pointing operation, the spacecraft solar-panel axis b_3 must align with the inertial n_2 direction.

The spacecraft operational modes are summarized as follows:

- **Power Mode:** active whenever the spacecraft is on the sunlit side of Mars,
- **Science Mode:** active during eclipse when GMO communication is unavailable,
- **Communication Mode:** active whenever the GMO spacecraft lies within the communication line-of-sight constraint.

The communication mode is activated when the angular separation between the LMO and GMO position vectors is less than 35° .

This operational logic prioritizes spacecraft power generation whenever solar illumination is available, while eclipse periods are allocated to science or communication operations.

Table 2.3: Mission Pointing Scenario Summary

Orbital Situation	Operational Mode
Spacecraft on sunlit Mars side	Power Mode
Spacecraft in eclipse and GMO visible	Communication Mode
Spacecraft in eclipse and GMO not visible	Science Mode

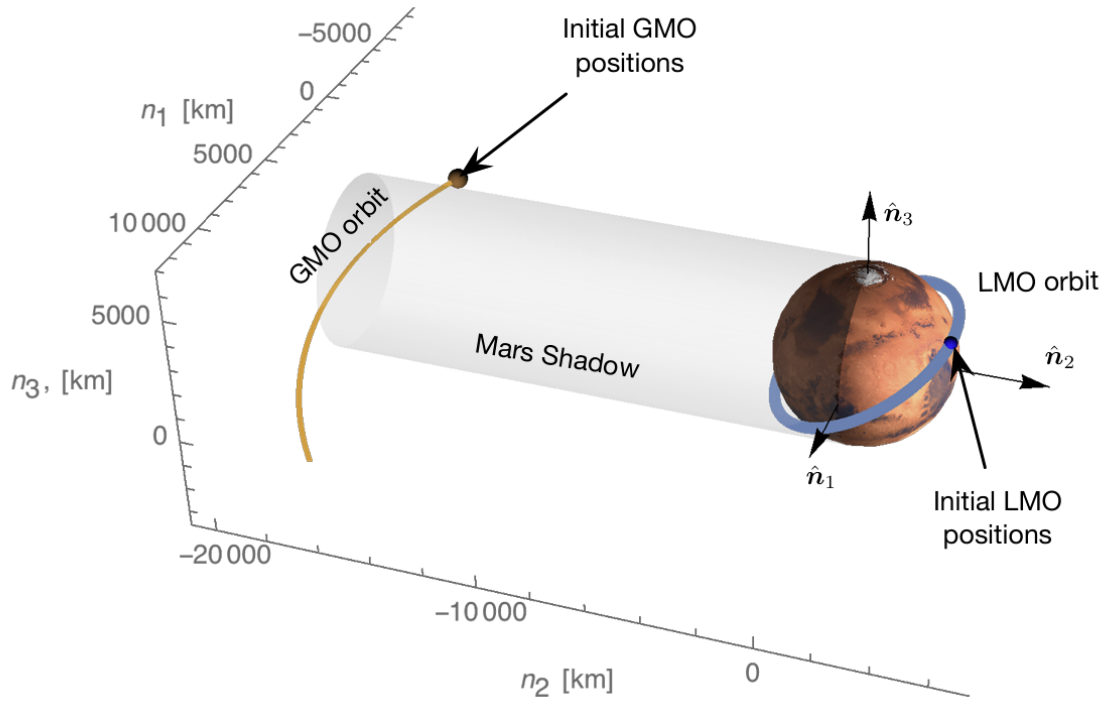


Figure 2.3: Mission Orbit Scenario Illustration

2.3 Spacecraft Attitude States

The spacecraft initial attitude is represented using Modified Rodrigues Parameters (MRPs), while the initial angular velocity is expressed in the spacecraft body frame.

The initial spacecraft attitude is defined as:

$$\sigma_{B/N}(t_0) = \begin{bmatrix} 0.3 \\ -0.4 \\ 0.5 \end{bmatrix}$$

The initial body angular velocity is:

$${}^B\omega_{B/N}(t_0) = \begin{bmatrix} 1.00 \\ 1.75 \\ -2.20 \end{bmatrix} \text{ deg/s}$$

The spacecraft inertia tensor is assumed diagonal, implying alignment between the body frame and principal inertia axes.

$${}^B[I] = \begin{bmatrix} 10 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 7.5 \end{bmatrix} \text{ kg}\cdot\text{m}^2$$

The spacecraft is assumed to be capable of generating arbitrary three-axis control torques through an onboard actuator system.

2.4 Attitude Control Law Overview

A proportional-derivative (PD) control law is used throughout this project to regulate spacecraft attitude motion:

$${}^B\mathbf{u} = -K\boldsymbol{\sigma}_{B/R} - P{}^B\boldsymbol{\omega}_{B/R} \quad (2.2)$$

where:

- $\boldsymbol{\sigma}_{B/R}$ is the attitude tracking error,
- ${}^B\boldsymbol{\omega}_{B/R}$ is the angular velocity tracking error,
- K and P are controller gain matrices.

The controller objective is to asymptotically reduce attitude and angular velocity tracking errors while ensuring stable closed-loop rotational behavior.

2.5 Implementation Methods

The project was implemented in MATLAB using modular functions for orbital propagation, reference-frame generation, attitude-error evaluation, and closed-loop numerical integration.

The simulation architecture was organized into four primary components:

- Orbit propagation functions to compute LMO and GMO inertial states as functions of time,
- Reference-frame functions to generate Sun-, nadir-, and GMO-pointing DCMs and angular velocities,
- Attitude utilities for MRP/DCM conversion and tracking-error evaluation,
- Mission simulation routines to propagate spacecraft dynamics while automatically switching between operational modes.

A fourth-order Runge–Kutta (RK4) integrator was used for numerical propagation of the attitude state. The modular structure simplified validation of individual tasks before integration into the complete mission scenario.

The final MATLAB results were then reconstructed in STK to visually verify spacecraft orientation and reference-frame tracking throughout the mission timeline.

Chapter 3

Task Descriptions and Solutions

3.1 Orbits

3.1.1 Task 1: Orbit Simulation

Assume the general orbit frame as $\mathcal{O} : \{\hat{i}_r, \hat{i}_\theta, \hat{i}_h\}$. Even though the general orbit frame and Hill frame basically use the same base vectors, their difference is found in their origin. $\mathcal{O}$ frame has its origin aligned with the origin of inertial frame $\mathcal{N}$ – which is the center of Mars. The Hill frame $\mathcal{H}$, however, is mounted on the spacecraft, therefore has the same origin as the body frame $\mathcal{B}$.

Both these frames are obtainable through a (3-1-3) Euler Angle rotation on the inertial frame, though the hill frame origin is also translated to the spacecraft.

Now, using the $\mathcal{O}$ frame, spacecraft position vector is expressed as:

$${}^{\mathcal{O}}\mathbf{r} = r\hat{i}_r$$

Both LMO and GMO orbits are circular, therefore having the same angular velocity $\dot{\theta}$ all over the orbit. Spacecraft velocity vector is expressed as:

$${}^{\mathcal{O}}\dot{\mathbf{r}} = r\dot{\theta}\hat{i}_\theta$$

The MATLAB function `orbitTheta()` takes time, angular velocity and initial angle to calculate the element θ using the following equation:

$$\theta_t = \dot{\theta}t + \theta_0$$

Taking the position and velocity vectors from the general orbit frame to the inertial frame is done using the DCM $[NO]$ which can be obtained having all orbital elements at a given time.

$$[ON] = M_3(\theta_t)M_1(i)M_3(\Omega)$$

$$[NO] = [ON]^T$$

$${}^{\mathcal{N}}\mathbf{r} = [NO] {}^{\mathcal{O}}\mathbf{r}$$

$${}^{\mathcal{N}}\dot{\mathbf{r}} = [NO] {}^{\mathcal{O}}\dot{\mathbf{r}}$$

This is achieved by the MATLAB function `orbit_state()`.

3.1.2 Task 2: Orbit Frame Orientation

The Hill-frame basis vectors were constructed directly from the inertial position and velocity vectors:

$${}^{\mathcal{N}}\hat{\mathbf{i}}_r = \frac{\mathbf{r}_{LMO}}{\|\mathbf{r}_{LMO}\|}$$

$${}^{\mathcal{N}}\hat{\mathbf{i}}_h = \frac{\mathbf{r}_{LMO} \times \dot{\mathbf{r}}_{LMO}}{\|\mathbf{r}_{LMO} \times \dot{\mathbf{r}}_{LMO}\|}$$

$${}^{\mathcal{N}}\hat{\mathbf{i}}_\theta = \hat{\mathbf{i}}_h \times \hat{\mathbf{i}}_r$$

To construct a DCM with the base vectors in either frames, we have:

$$[FB] = \begin{bmatrix} {}^{\mathcal{F}}\hat{\mathbf{b}}_1 & {}^{\mathcal{F}}\hat{\mathbf{b}}_2 & {}^{\mathcal{F}}\hat{\mathbf{b}}_3 \end{bmatrix} = \begin{bmatrix} {}^{\mathcal{B}}\hat{\mathbf{f}}_1^T \\ {}^{\mathcal{B}}\hat{\mathbf{f}}_2^T \\ {}^{\mathcal{B}}\hat{\mathbf{f}}_3^T \end{bmatrix}$$

Therefore, to find $[HN]$ we will have:

$$[HN] = \begin{bmatrix} {}^{\mathcal{N}}\hat{\mathbf{i}}_r^T \\ {}^{\mathcal{N}}\hat{\mathbf{i}}_\theta^T \\ {}^{\mathcal{N}}\hat{\mathbf{i}}_h^T \end{bmatrix}$$

A MATLAB function named `orbit_frame_dcm(t)` was implemented to evaluate the time-varying Hill-frame orientation.

3.2 Reference Frame Orientation

3.2.1 Task 3: Sun-Pointing Reference Frame Orientation

To point the spacecraft solar panels axis $\hat{\mathbf{b}}_3$ at the Sun, the reference frame $\mathcal{R}_s$ must be chosen such that $\hat{\mathbf{r}}_3$ axis points in the Sun direction ($\hat{\mathbf{n}}_2$ in the scenario). To define the

frame completely, the first axis $\hat{\mathbf{r}}_1$ is assumed to point in $-\hat{\mathbf{n}}_1$ direction. Therefore, the base vectors will be:

$$\begin{aligned}\hat{\mathbf{r}}_1 &= -\hat{\mathbf{n}}_1 \\ \hat{\mathbf{r}}_3 &= \hat{\mathbf{n}}_2 \\ \hat{\mathbf{r}}_2 &= \hat{\mathbf{r}}_3 \times \hat{\mathbf{r}}_1 = -(\hat{\mathbf{n}}_2 \times \hat{\mathbf{n}}_1) = \hat{\mathbf{n}}_3\end{aligned}$$

Similar to the previous task, the analytical expression for the DCM can be obtained using the base vectors:

$$[R_s N] = \begin{bmatrix} \mathcal{N} \hat{\mathbf{r}}_1^T \\ \mathcal{N} \hat{\mathbf{r}}_2^T \\ \mathcal{N} \hat{\mathbf{r}}_3^T \end{bmatrix} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

The resulting reference-frame DCM can also be obtained using a 3-1-3 Euler-angle of $(\pi, \frac{\pi}{2}, 0)$ rotation on it. MATLAB function `RsN_DCM()` serves this purpose.

Since the base vectors of this frame are just a fixed permutation of inertial frame basis vectors, the frame remains fixed relative to the inertial frame, thus:

$$\boldsymbol{\omega}_{R_s/N} = 0$$

3.2.2 Task 4: Nadir-Pointing Reference Frame Orientation

To point the spacecraft sensor platform axis $\hat{\mathbf{b}}_1$ towards the center of Mars (nadir direction), the reference frame $\mathcal{R}_n$ must be chosen such that $\hat{\mathbf{r}}_1$ axis points towards the planet. To completely describe the frame, we assume the second axis $\hat{\mathbf{r}}_2$ points in velocity direction $\hat{\mathbf{i}}_\theta$. Therefore, we'll have:

$$\begin{aligned}\hat{\mathbf{r}}_1 &= -\hat{\mathbf{i}}_r \\ \hat{\mathbf{r}}_2 &= \hat{\mathbf{i}}_\theta \\ \hat{\mathbf{r}}_3 &= \hat{\mathbf{r}}_1 \times \hat{\mathbf{r}}_2 = -\hat{\mathbf{i}}_h\end{aligned}$$

This frame can be obtained using a (3-1-3) Euler Angles rotation on the Hill frame with angles $(\pi, \pi, 0)$.

$$[R_n H] = M_1(\pi)M_3(\pi)$$

We also had previously obtained $[HN]$ in task 2 as a function of time. So,

$$[R_n N](t) = [R_n H][HN](t)$$

In MATLAB, this task is done by function `RnN_DCM()`.

Since this frame is a re-arrangement of the Hill frame, it also moves with it, thus:

$$\boldsymbol{\omega}_{R_n/N} = \boldsymbol{\omega}_{H/N} = \dot{\theta} \hat{\mathbf{i}}_\theta$$

3.2.3 Task 5: GMO-Pointing Reference Frame Orientation

To point the nano-satellite communication platform axis $\hat{\mathbf{b}}_1$ towards the GMO mother spacecraft, the communication reference frame $\mathcal{R}_c$ must be chosen such that $-\hat{\mathbf{r}}_1$ axis points towards the GMO satellite location, defining $\Delta \mathbf{r} = \mathbf{r}_{GMO} - \mathbf{r}_{LMO}$. So,

$$\hat{\mathbf{r}}_1 = -\frac{\Delta \mathbf{r}}{\|\Delta \mathbf{r}\|}$$

Further, to fully define a three-dimensional reference frame, assume the second axis is defined as

$$\hat{\mathbf{r}}_2 = \frac{\Delta \mathbf{r} \times \hat{\mathbf{n}}_3}{\|\Delta \mathbf{r} \times \hat{\mathbf{n}}_3\|}$$

The third basis vector is then obtained from

$$\hat{\mathbf{r}}_3 = \hat{\mathbf{r}}_1 \times \hat{\mathbf{r}}_2$$

Having all three base vectors in the inertial frame, we have:

$$[R_c N] = \begin{bmatrix} \mathcal{N} \hat{\mathbf{r}}_1^T \\ \mathcal{N} \hat{\mathbf{r}}_2^T \\ \mathcal{N} \hat{\mathbf{r}}_3^T \end{bmatrix}$$

The MATLAB function `RcN_DCM(t)` is written to deliver this task.

To derive the relative angular velocity ${}^{\mathcal{N}}\boldsymbol{\omega}_{R_c/N}$, we'll use an equation (3.27) in the textbook [1] :

$$[\dot{C}] = -[\tilde{\boldsymbol{\omega}}][C] \quad (3.1)$$

Because the reference-frame motion varies smoothly over time, finite-difference differentiation provides a sufficiently accurate approximation of the frame angular velocity. Therefore,

$$[\tilde{\boldsymbol{\omega}}_{R_c/N}] = \begin{bmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{bmatrix} = -[R_c \dot{N}][R_c N]^T \quad (3.2)$$

To derive the rate of a DCM numerically, we'll have:

$$[R_c \dot{N}] \approx \frac{[R_c N]_{t+dt} - [R_c N]_t}{dt} \quad (3.3)$$

This algorithm is implemented in MATLAB function `RcN_omega(t)`. The choice of different `dt` will result in different precisions in finding the angular velocity.

3.3 Attitude Evaluation and Simulator

3.3.1 Task 6: Attitude Error Evaluation

Any attitude feedback control algorithm requires the ability to compute the attitude and angular velocity tracking errors of the current body frame $\mathcal{B}$ relative to the reference frame $\mathcal{R}$. To find a set of tracking errors $\begin{bmatrix} \sigma_{B/R} \\ {}^{\mathcal{B}}\omega_{B/R} \end{bmatrix}$, we have the known variables $\sigma_{B/N}$, ${}^{\mathcal{B}}\omega_{B/N}$, as well as the desired reference frame DCM $[RN]$ and frame rate ${}^{\mathcal{N}}\omega_{R/N}$. We first construct the DCM $[BN]$ which can be obtained from $\sigma_{B/N}$ using equation (3.152):

$$[C] = [I_{3 \times 3}] + \frac{8[\tilde{\sigma}]^2 - 4(1 - \sigma^2)[\tilde{\sigma}]}{(1 + \sigma^2)^2} \quad (3.4)$$

The inverse transformation—converting a DCM to its MRP form—is formulated in equation (3.153):

$$\begin{aligned} [\tilde{\sigma}] &= \frac{[C]^T - [C]}{\zeta(\zeta + 2)} \\ \zeta &= \sqrt{\text{trace}([C]) + 1} \end{aligned} \quad (3.5)$$

These two forms are implemented in MATLAB function `sigmaToDCM()` and `DCMToSigma()` respectively.

Now, the roadmap to finding the tracking errors would be:

$$\begin{aligned} \sigma_{B/N} &\rightarrow [BN] \\ [BR] &= [BN][NR] = [BN][RN]^T \\ [BR] &\rightarrow \sigma_{B/R} \\ {}^{\mathcal{B}}\omega_{B/R} &= {}^{\mathcal{B}}\omega_{B/N} - {}^{\mathcal{B}}\omega_{R/N} = {}^{\mathcal{B}}\omega_{B/N} - [BN]{}^{\mathcal{N}}\omega_{R/N} \end{aligned}$$

A MATLAB function named `attitudeErrorEval()` was implemented to evaluate both attitude and angular velocity tracking errors.

3.3.2 Task 7: Numerical Attitude Simulator

In order to integrate the attitude we need a numerical integrator. The LMO and GMO orbit satellite positions are already calculated in earlier tasks. Let the propagated attitude

state $\mathbf{X}$ be

$$\mathbf{X} = \begin{bmatrix} \boldsymbol{\sigma}_{B/N} \\ {}^{\mathcal{B}}\boldsymbol{\omega}_{B/N} \end{bmatrix}$$

Assuming that the spacecraft is rigid and there's no external torque acting on it, its dynamics obey

$$[I]\dot{\boldsymbol{\omega}}_{B/N} = -[\tilde{\boldsymbol{\omega}}_{B/N}][I]\boldsymbol{\omega}_{B/N} + \mathbf{u} \quad (3.6)$$

where $\mathbf{u}$ is the external control torque vector.

For this task, a loop integrator was written in MATLAB. The control torques ${}^{\mathcal{B}}\mathbf{u}_1 = 0$ and ${}^{\mathcal{B}}\mathbf{u}_2 = (0.01, -0.01, 0.02)$ were applied respectively and the following were computed:

- Attitude MRP $\boldsymbol{\sigma}_{B/N}$
- Angular velocity $\boldsymbol{\omega}_{B/N}$
- Angular momentum vector $\mathbf{H} = [I]\boldsymbol{\omega}_{B/N}$ in both $\mathcal{N}$ and $\mathcal{B}$ frames
- Rotational kinetic energy $T = \frac{1}{2}\boldsymbol{\omega}_{B/N}^T [I]\boldsymbol{\omega}_{B/N}$

The results of both simulations are plotted in Figure 3.1. As is seen, after applying the control torque $\mathbf{u}_2$ on the spacecraft, its rotational kinetic energy increases significantly (from approximately 9 mJ in uncontrolled mode (Figure 3.1a) to more than 2.5 J when controlled (Figure 3.1b)). This is verified by spacecraft angular velocity, whose 1st and 2nd components start to oscillate in controlled mode while its 3rd goes through a rapid, non-stop increase (Figure 3.1d).

The attitude-state histories further confirm this behavior. The spacecraft exhibits unstable rotational behavior when the control torque $\mathbf{u}_2$ is applied to it (Figure 3.1f). These results indicate that the applied torque does not stabilize the spacecraft attitude dynamics.

3.4 Completing the Mission

3.4.1 Task 8: Sun Pointing Control

The PD gains were selected using the linearized closed-loop attitude dynamics together with a worst-case inertia design approach. Since scalar gains were used for all spacecraft axes, the gains were chosen based on the maximum and minimum principal moments of inertia to guarantee that all rotational modes satisfy the required 120-second decay time constant.

$$I_{max} = 10, I_{min} = 5$$

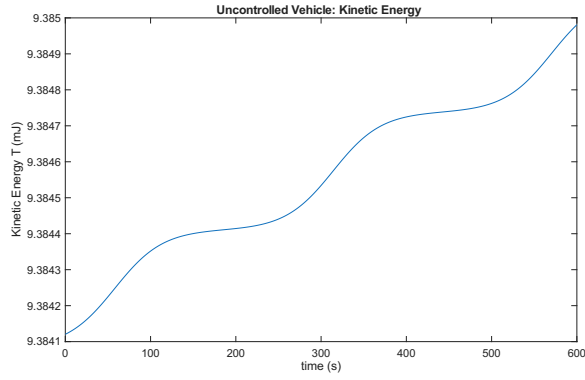
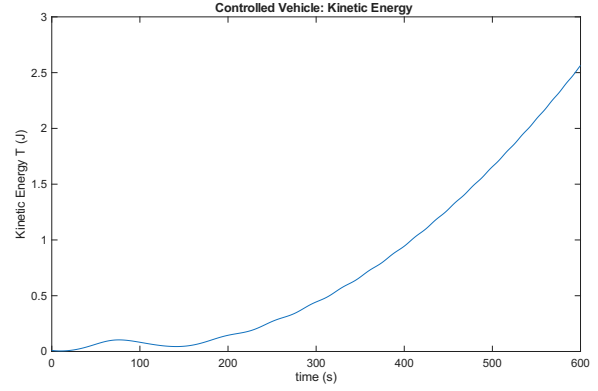
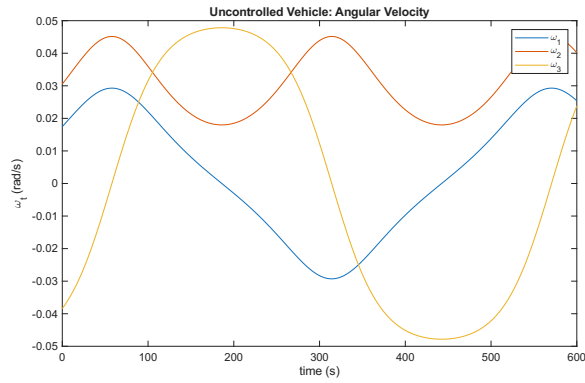
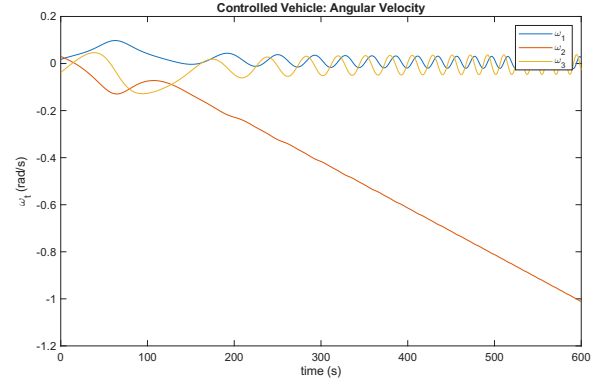
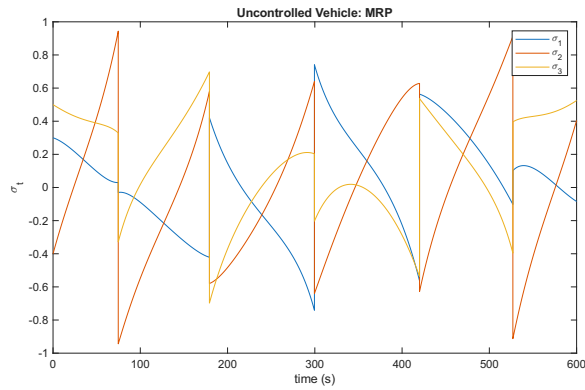
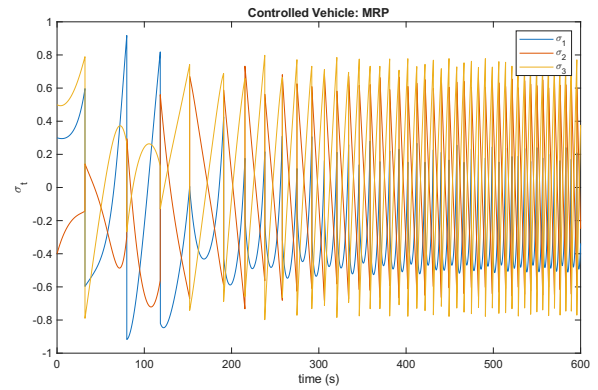
(a) u_1 applied: T (b) u_2 applied: T (c) u_1 applied: $\omega_{B/N}$ (d) u_2 applied: $\omega_{B/N}$ (e) u_1 applied: $\sigma_{B/N}$ (f) u_2 applied: $\sigma_{B/N}$

Figure 3.1: Results of the simulations in task 7

$$\tau = 120s$$

$$K = \frac{4I_{max}^2}{\tau^2 I_{min}}, P = \frac{2I_{max}}{\tau} \quad (3.7)$$

The control torque is defined as:

$${}^B\mathbf{u} = -K\boldsymbol{\sigma}_{B/R} - P{}^B\boldsymbol{\omega}_{B/R}$$

An RK4 integrator was developed to perform numerical propagation of the spacecraft attitude dynamics used by the MATLAB function `simulate_mode()`. The results of Sun pointing simulation are evident in Figure 3.2.

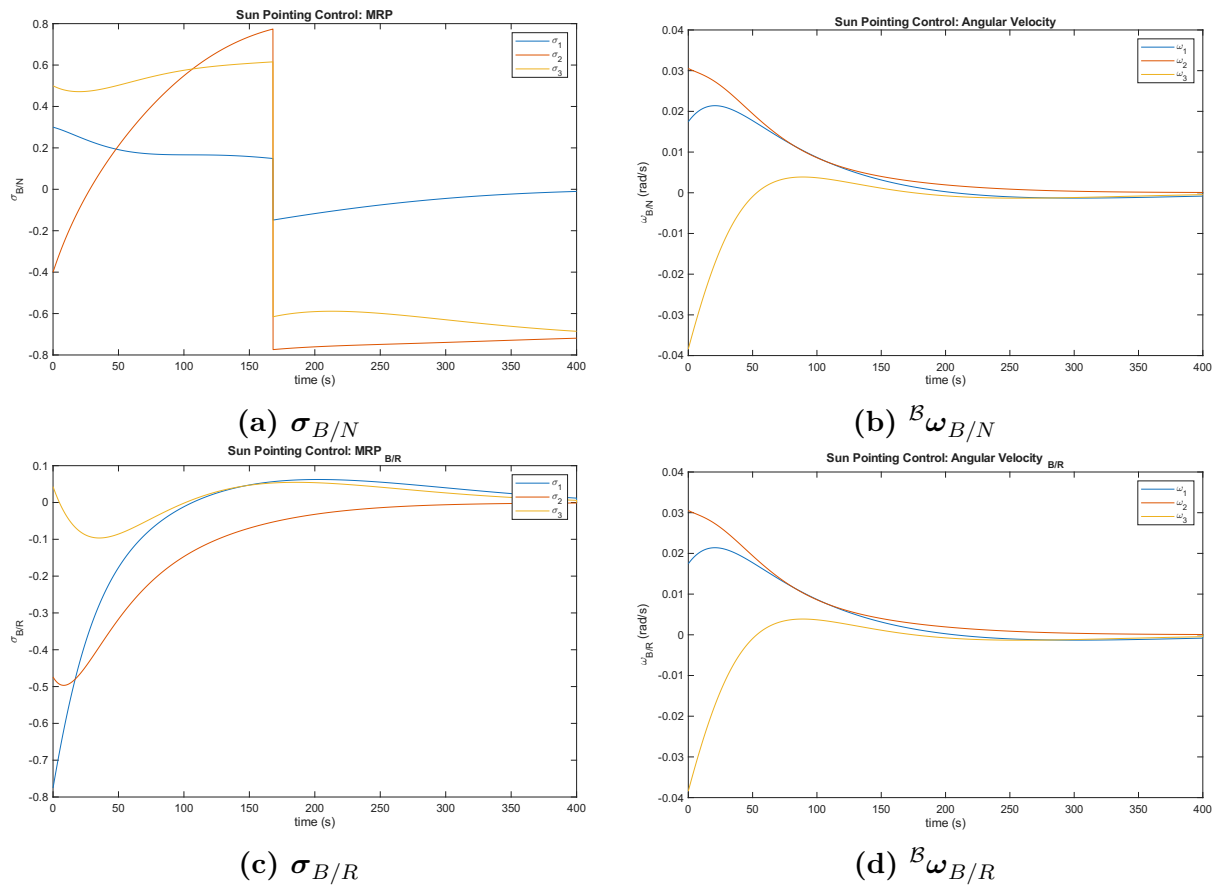


Figure 3.2: Results of Sun pointing control

The attitude and angular-velocity tracking errors converge toward zero, demonstrating successful stabilization of the spacecraft about the inertially fixed Sun-pointing reference frame. The body rates are progressively damped by the PD controller, resulting in effective detumbling and accurate alignment of the solar-panel axis with the desired Sun direction.

3.4.2 Task 9: Nadir Pointing Control

The same PD control law developed for Sun-pointing operation was applied to the nadir-pointing reference frame. Unlike the Sun-pointing case, however, the nadir frame rotates continuously with the orbital motion of the spacecraft and therefore represents a time-varying attitude-tracking problem.

The results shown in Figure 3.3 demonstrate successful convergence of both the attitude and angular-velocity tracking errors. Following an initial transient response, the MRP tracking-error components decrease steadily while the angular-velocity tracking errors converge to zero. The spacecraft simultaneously detumbles and aligns its sensor axis with the Mars-facing nadir direction.

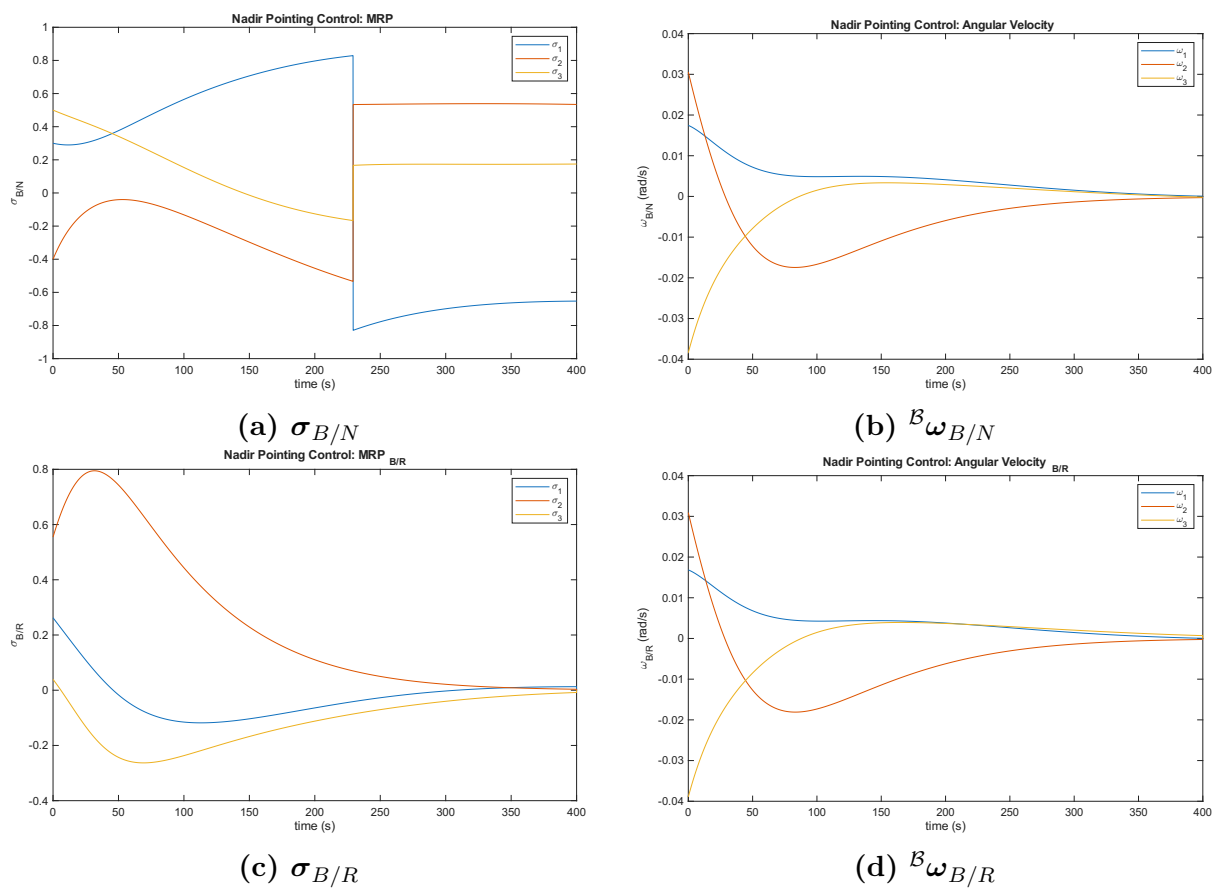


Figure 3.3: Results of Nadir pointing control

Compared with the inertially fixed Sun-pointing mode, convergence occurs more gradually because the controller must continuously track the rotating reference frame while reducing the initial attitude error. Nevertheless, the response remains stable throughout the simulation and exhibits behavior consistent with the controller design objective of a 120-second closed-loop decay time constant.

3.4.3 Task 10: GMO Pointing Control

The same control architecture was next evaluated for GMO-pointing operation, where the spacecraft communication antenna is required to maintain alignment with the geostationary Mars orbiter. This scenario represents one of the most dynamically demanding tracking problems considered in the project because the reference frame orientation depends on the continuously changing relative geometry between the LMO and GMO spacecraft.

Figure 3.4 shows that the controller successfully drives both the attitude and angular-velocity tracking errors toward zero. The transient response immediately following the simulation start is rapidly damped, and the tracking errors remain bounded throughout the maneuver. As in the nadir-pointing case, the spacecraft must track a rotating reference frame rather than a fixed inertial orientation.

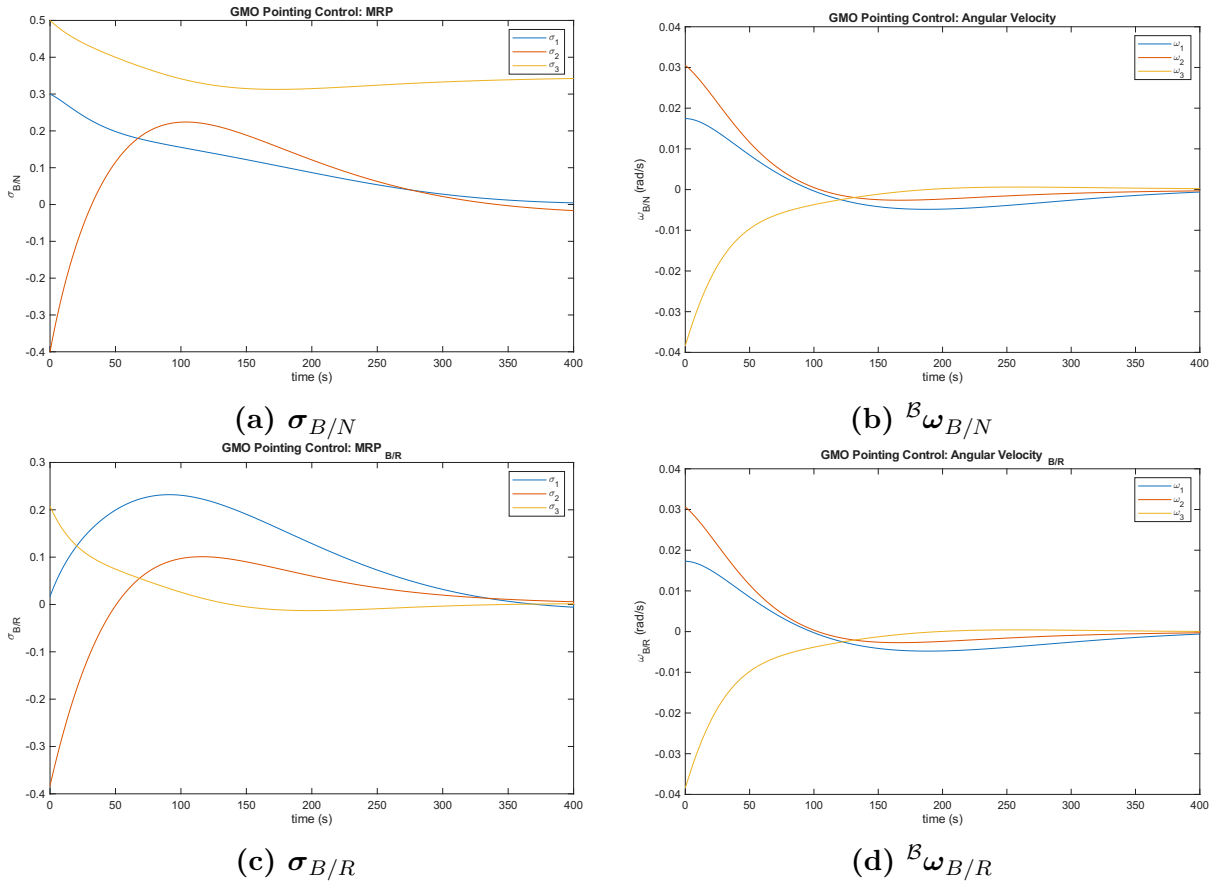


Figure 3.4: Results of GMO pointing control

The results demonstrate that the implemented PD controller provides sufficient authority to maintain communication-pointing requirements despite the time-varying nature of the reference attitude. The observed convergence behavior is comparable to that predicted by the gain-selection methodology and confirms successful tracking of the GMO communication target.

3.4.4 Task 11: Mission Scenario Simulation

A complete mission simulation was developed to automatically switch between spacecraft operational modes according to mission conditions.

The mission logic was implemented as follows:

- If the spacecraft was illuminated by the Sun, Sun-pointing mode was activated,
- If the spacecraft was in eclipse and the GMO spacecraft was visible, communication mode was activated,
- Otherwise, nadir-pointing science mode was selected.

The resulting mission sequence consists of alternating Sun-pointing, nadir-pointing, and GMO-pointing intervals determined by orbital illumination and communication geometry. Figures 3.5 and 3.6 present the complete spacecraft attitude and angular-velocity histories over the 6500-second mission duration.

Each operational-mode transition introduces a discontinuous change in the commanded reference frame, producing temporary increases in the attitude and angular-velocity tracking errors. These transient responses are expected and represent the spacecraft reorienting toward a new mission objective. In all cases, the controller successfully damps the resulting errors and restores tracking performance without loss of stability.

The simulation demonstrates that the spacecraft can autonomously execute mission-level decision logic while maintaining closed-loop attitude control. The controller remains effective throughout repeated mode transitions and preserves acceptable pointing performance across all operational phases.

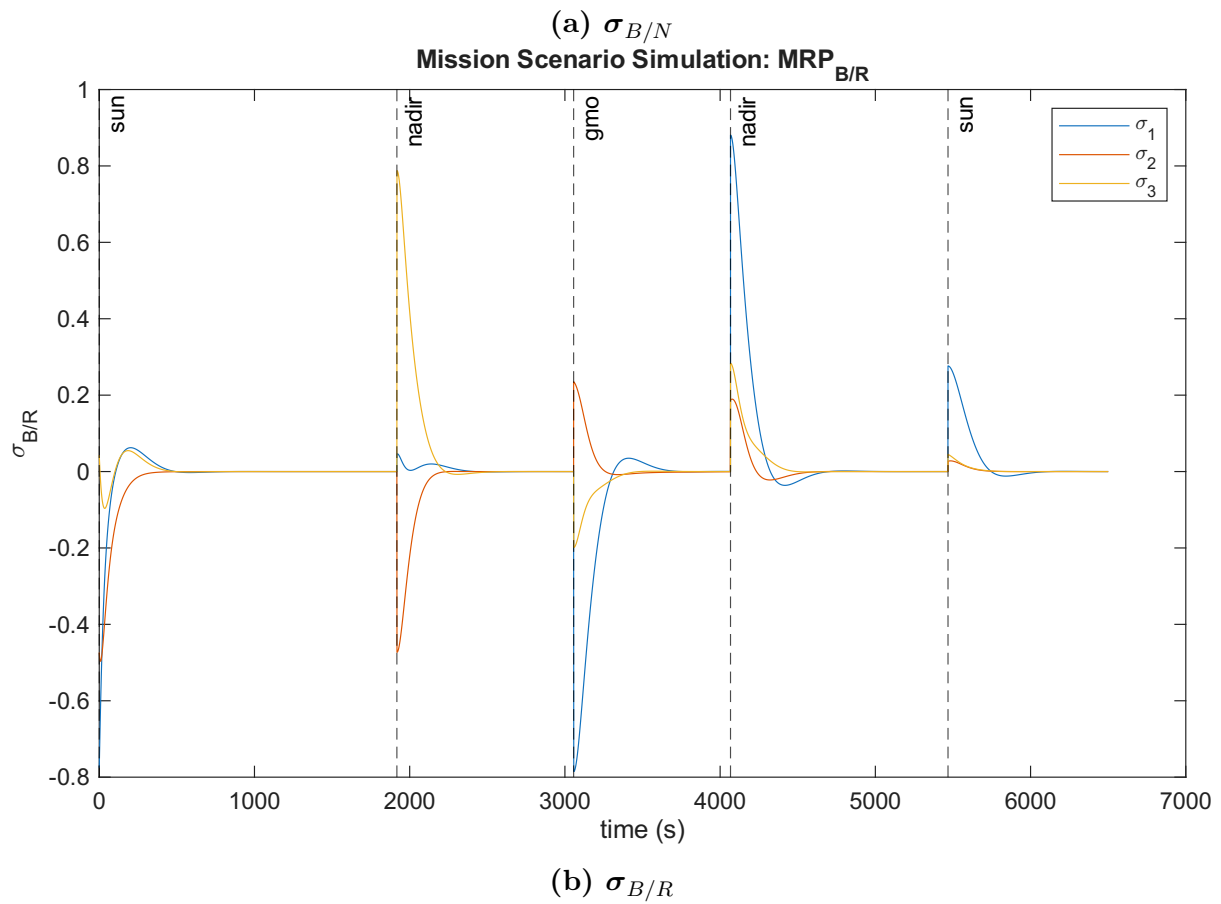
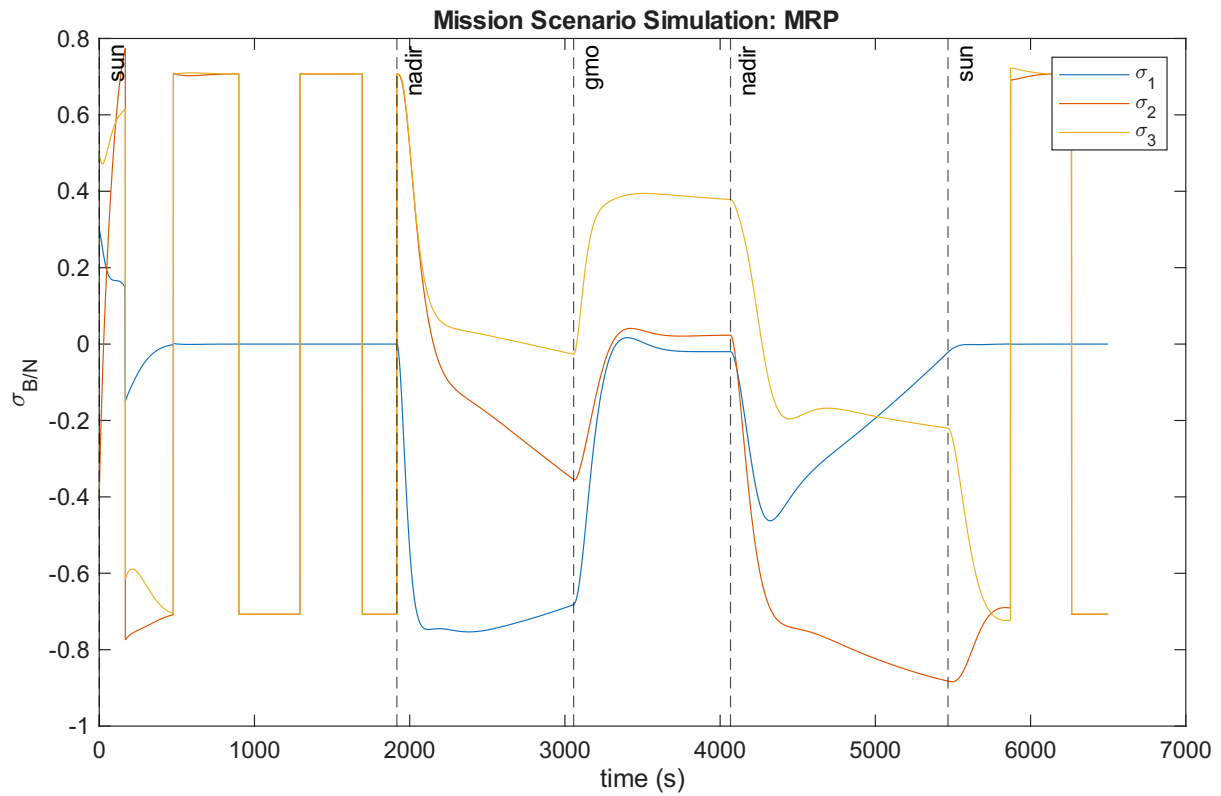


Figure 3.5: Full mission spacecraft attitude state

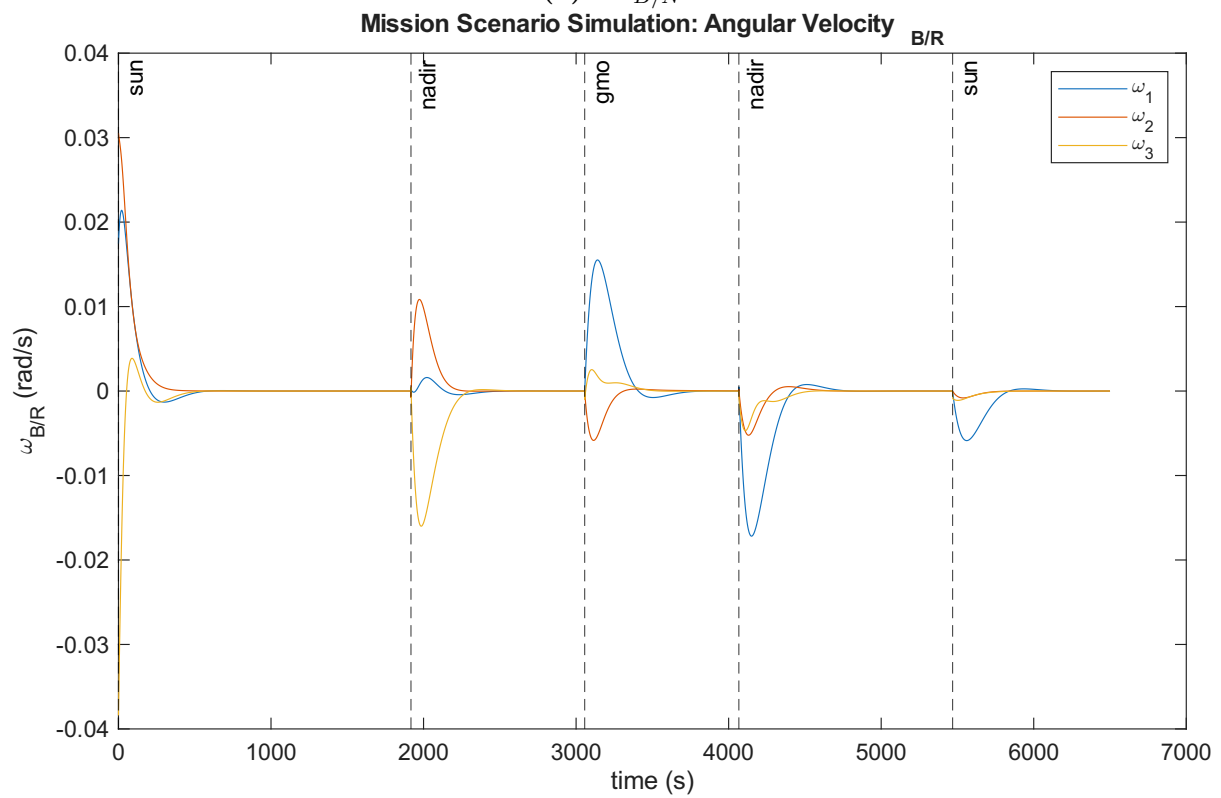
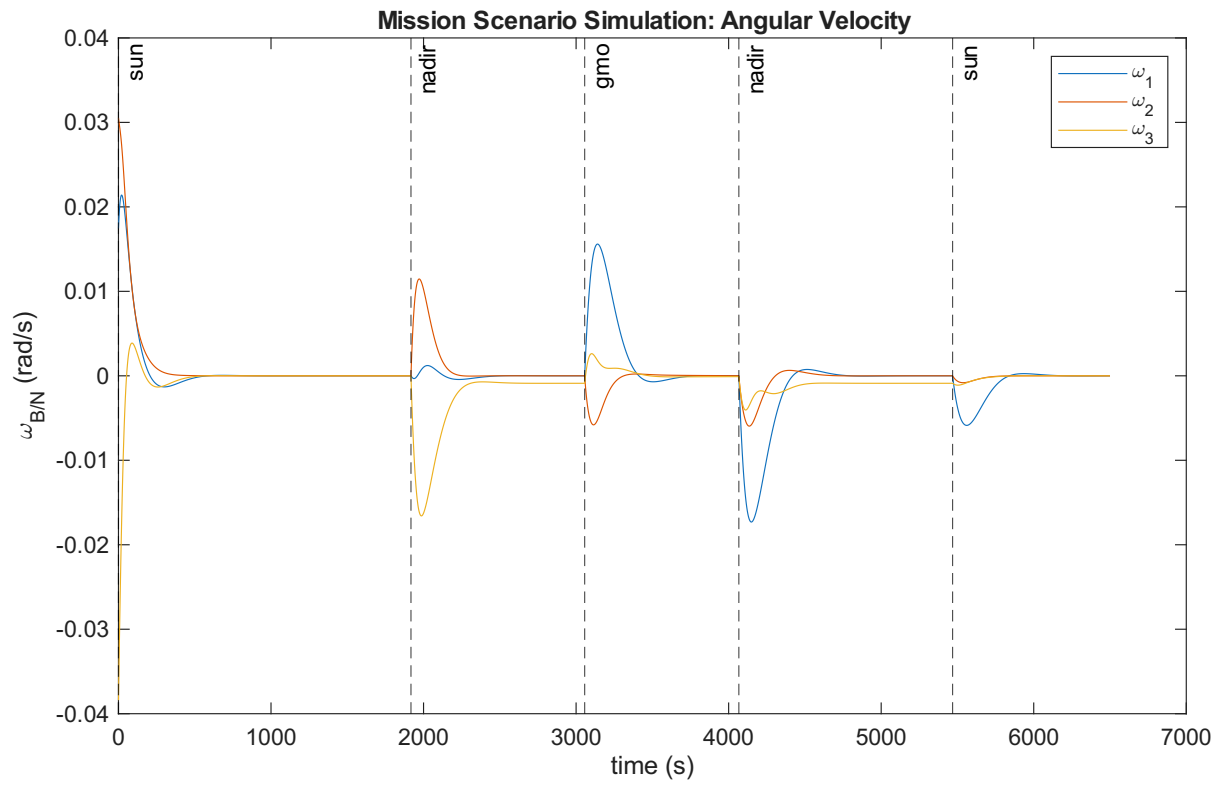


Figure 3.6: Full mission spacecraft angular velocity

3.5 Simulation Results and Discussion

The complete mission simulation demonstrates that the spacecraft attitude-control architecture successfully executes autonomous transitions between multiple operational pointing modes while maintaining stable closed-loop behavior throughout the 6500-second mission timeline.

Unlike the single-mode simulations of Tasks 8–10, the mission scenario requires continuous switching between Sun-pointing, nadir-pointing, and GMO-pointing reference frames according to orbital geometry and mission constraints. Consequently, the controller must not only regulate the spacecraft attitude but also track time-varying reference commands and recover from abrupt changes in the desired orientation.

The results presented in Figures 3.5 and 3.6 show that the spacecraft remains stable throughout all mission phases. Following each mode transition, temporary increases in attitude and angular-velocity tracking errors are observed due to the discontinuous change in the commanded reference frame. These transient responses are rapidly damped by the PD controller and do not lead to sustained oscillations or instability.

The controller gains were selected using a worst-case inertia design corresponding to a desired closed-loop decay time constant of approximately 120 seconds. Examination of the tracking-error histories confirms behavior consistent with this design objective. The dominant attitude-error components decrease significantly within the first few hundred seconds following a mode transition, and the angular-velocity tracking errors converge toward zero over a similar time scale. Most transient activity is removed within approximately three time constants (≈ 360 s), which agrees well with the analytical gain-selection methodology.

Sun-pointing operation exhibits the fastest convergence because the associated reference frame is fixed in inertial space. Nadir-pointing and GMO-pointing modes require tracking rotating reference frames and therefore maintain small residual tracking errors even after the initial transient has decayed. Nevertheless, the tracking errors remain bounded and continue to converge toward their commanded trajectories throughout each operational segment.

Overall, the simulation results demonstrate that the implemented PD control law provides effective detumbling, stable reference tracking, and reliable autonomous mode transitions. The spacecraft successfully satisfies all mission pointing objectives while maintaining acceptable attitude-tracking performance across the complete mission timeline.

Chapter 4

STK Visualization

To complement the numerical MATLAB simulations, the complete mission scenario was reconstructed in Systems Tool Kit (STK). The same orbital geometry, mission logic, and spacecraft attitude histories generated in MATLAB were imported into STK to provide independent three-dimensional validation of the spacecraft pointing behavior throughout the mission timeline.

The visualization provides an independent assessment of the spacecraft pointing performance and serves as a qualitative validation of the numerical results. Particular attention was given to mode-transition events and the spacecraft's ability to align with the commanded Sun-pointing, nadir-pointing, and GMO-pointing reference frames.

Figure 4.1 illustrates the Mars-centered inertial reference frame together with the LMO and GMO spacecraft orbits used throughout the mission simulation. Figure 4.2 shows the mission geometry and reference vectors used to define the operational pointing modes. The following sections examine each mission phase individually and compare the visual observations with the numerical tracking results presented in Chapter 3.

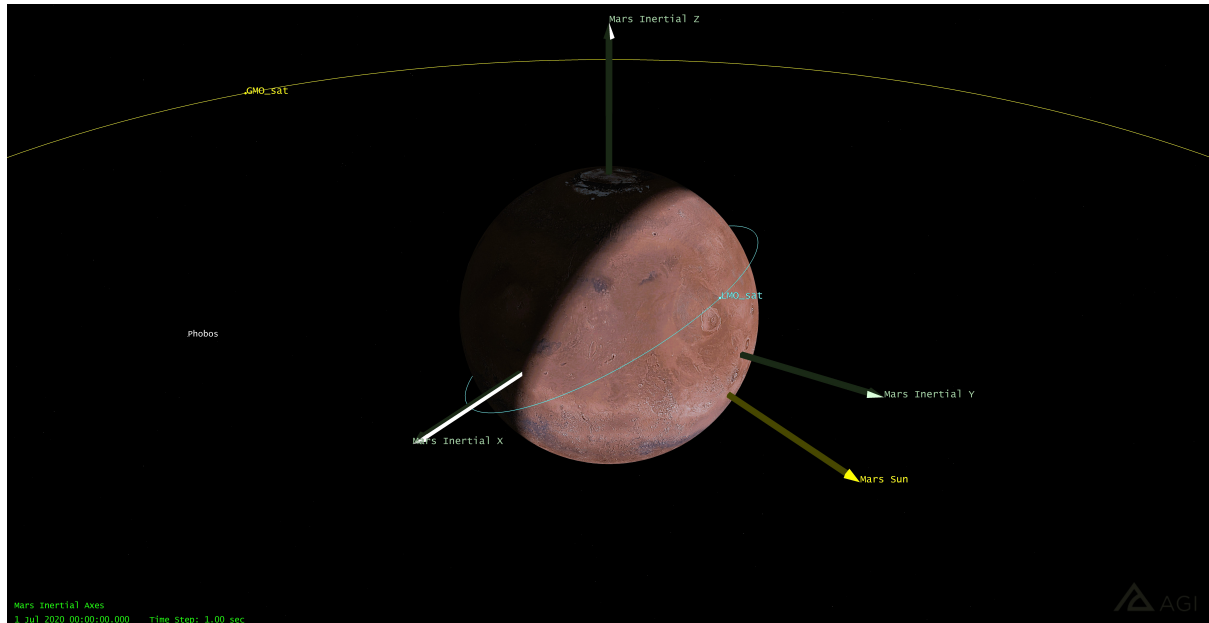


Figure 4.1: Illustration of the central body inertial frame with sun vector (yellow) alongside the spacecrafts' orbits with the blue orbit indicating LMO and the yellow orbit, GMO. The figure depicts the inertial orbital positions of the spacecrafts.

An assumption adopted in the project considered the Sun to be at an effectively infinite distance, with inertial basis vector $\hat{n}_2$ (in this case, **Mars Inertial Y**) aligned toward it. In real world simulation, however, the date and time 1 July 2020 was the closest to this assumption.

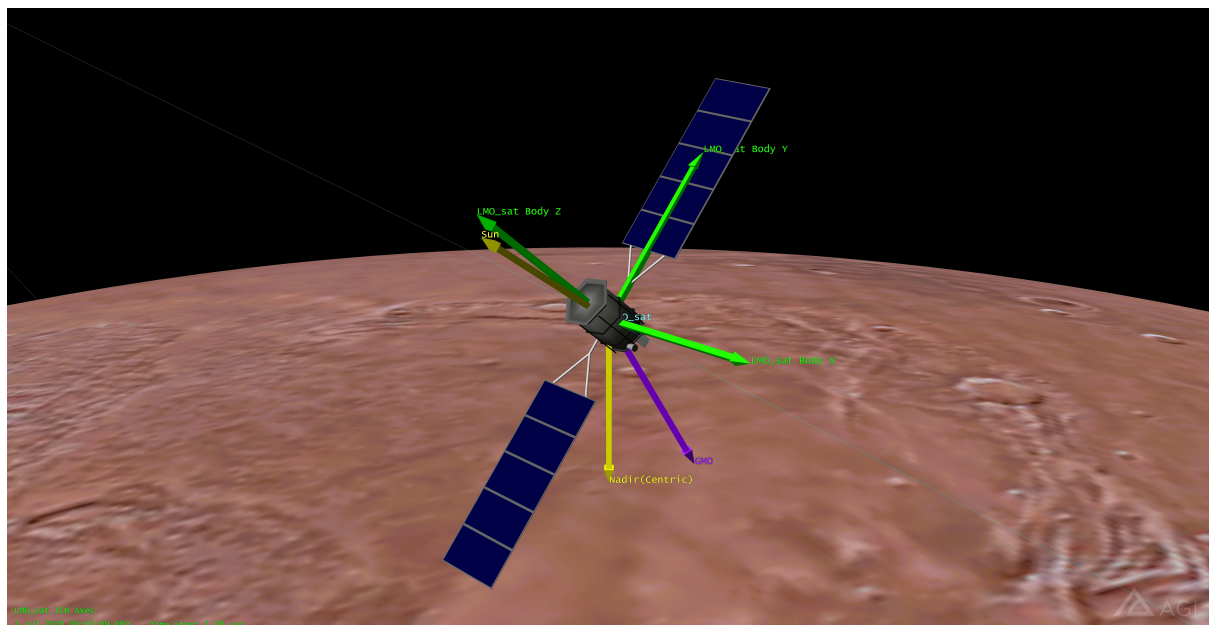


Figure 4.2: LMO spacecraft inertial position and mission reference vectors in STK

4.1 Sun Pointing Interval

During Sun-pointing intervals, the spacecraft solar-panel axis remains aligned with the prescribed inertial Sun direction throughout the operational segment. The STK visualization confirms the numerical simulation results by showing stable geometric alignment between the spacecraft body frame and the Sun-pointing reference frame. Only brief re-orientation maneuvers are observed immediately following mode transitions, after which the spacecraft maintains the required power-generation attitude.

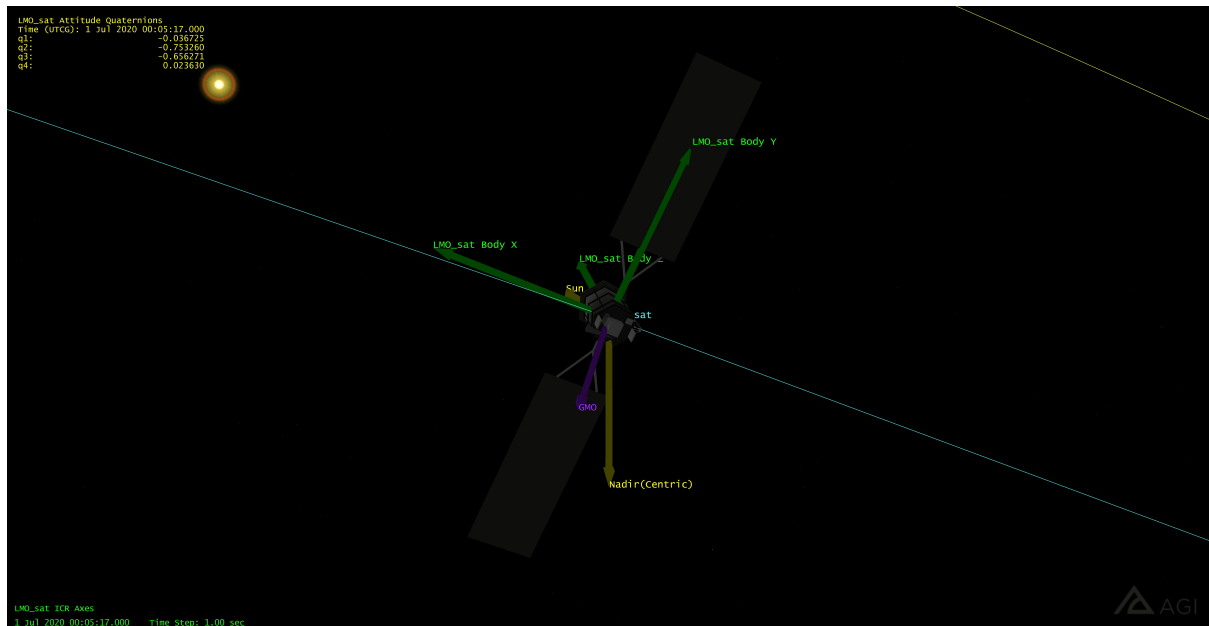


Figure 4.3: STK Sun Pointing Mode

4.2 Nadir Pointing Interval

During the nadir-pointing intervals, the spacecraft successfully reorients such that the sensor-aligned body axis tracks the Mars-facing reference direction. The STK visualization confirms the behavior observed in the numerical simulations, where the attitude and angular-velocity tracking errors decrease following each transition into science mode.

Compared with the Sun-pointing case, the nadir reference frame rotates continuously with the orbital motion of the spacecraft. Consequently, the controller must simultaneously remove the initial attitude error and track a time-varying reference orientation. This produces larger transient responses immediately after mode transitions and a slower convergence process than that observed for the inertially fixed Sun-pointing reference frame.

The observed response remains consistent with the controller design objectives. The spacecraft rapidly reduces its initial tracking error, maintains stable rotational behavior,

and achieves visually correct Mars-pointing orientation throughout the science operations interval. No evidence of instability, divergence, or persistent pointing offset is observed in either the MATLAB simulations or the STK reconstruction.

The agreement between the numerical results and the STK visualization provides strong validation of the nadir-pointing reference-frame implementation and confirms successful tracking of the rotating Mars-pointing attitude command.

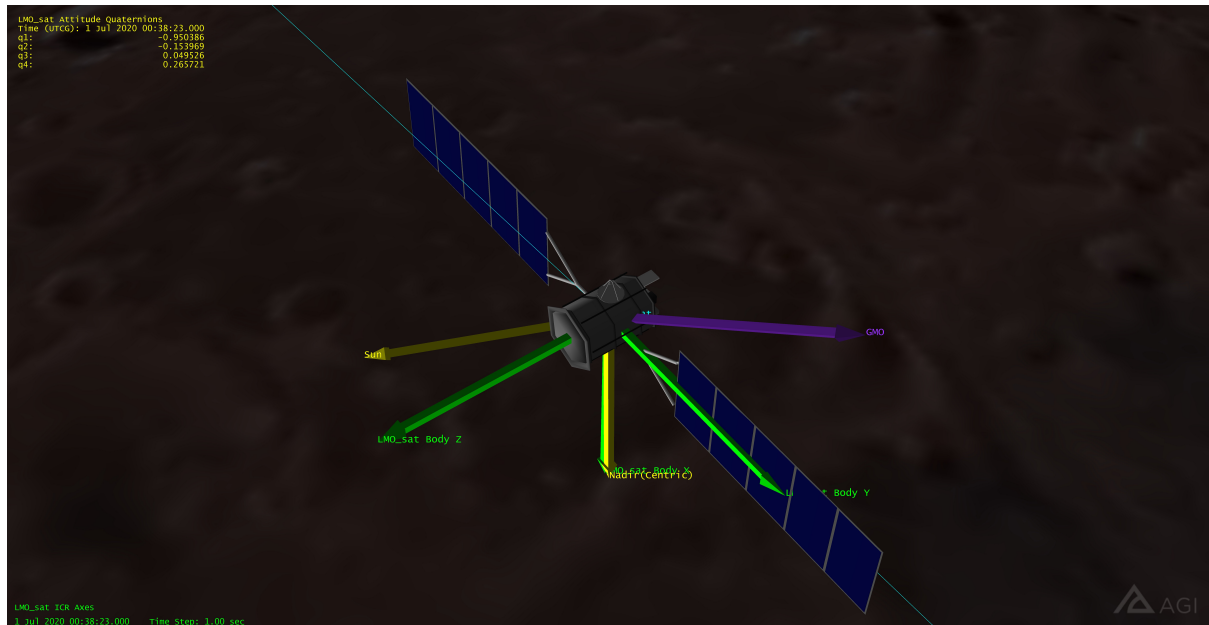


Figure 4.4: STK Nadir Pointing Mode

4.3 GMO Pointing Interval

During GMO-pointing operation, the spacecraft successfully reorients toward the communication target and maintains a visually consistent line-of-sight throughout the segment. The numerical MRP response shows transient behavior immediately after switching reference frames followed by convergence to bounded values, which agrees with the visual observation in STK.

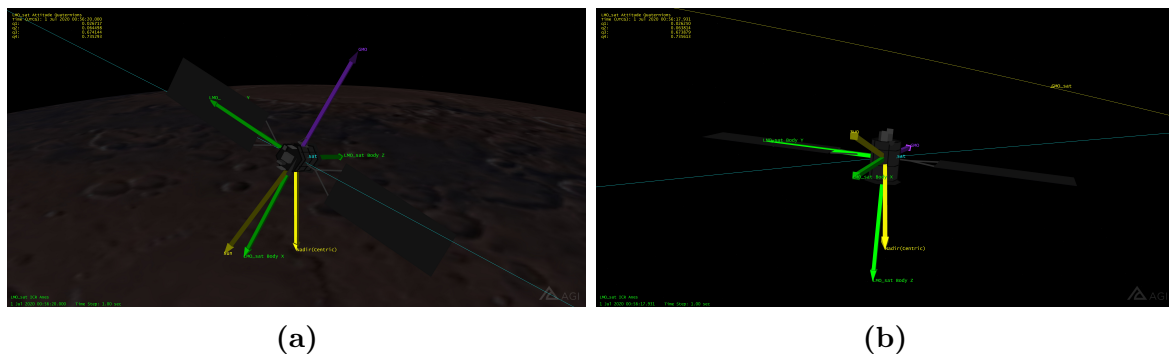


Figure 4.5: STK GMO Pointing Mode

Unlike the Sun-pointing mode, the GMO-pointing reference frame is generated from the instantaneous relative position vector between the LMO and GMO spacecraft. Consequently, successful tracking requires continuous adjustment of the spacecraft attitude as the orbital geometry evolves. The STK visualization confirms that the communication antenna remains properly oriented toward the GMO spacecraft throughout the communication interval, validating both the reference-frame formulation and the closed-loop tracking performance observed in MATLAB.

4.4 Mission-level Validation Summary

The STK reconstruction provides independent visual confirmation of the numerical simulation results obtained in MATLAB. Throughout the mission timeline, the spacecraft remains dynamically stable, successfully transitions between operational modes, and maintains the required pointing objectives for power generation, science observations, and communications.

The visual behavior observed during Sun-pointing, nadir-pointing, and GMO-pointing intervals is consistent with the corresponding attitude-error histories presented in Chapter 3. In all three modes, the spacecraft reorients toward the commanded reference frame following mode transitions and subsequently maintains the desired pointing geometry.

Particularly important is the agreement between the numerical and visual results during nadir-pointing operation. The STK reconstruction confirms that the spacecraft achieves the required Mars-facing orientation and that the apparent transient errors observed immediately after mode transitions correspond to normal closed-loop convergence behavior rather than a reference-frame implementation error.

Overall, the combined MATLAB and STK analysis validates the complete attitude determination and control framework developed in this project. The agreement between analytical predictions, numerical simulation, and visual reconstruction provides confidence in both the reference-frame formulations and the implemented control architecture.

Conclusion

This project successfully developed and validated a complete attitude dynamics and control system for a nano-satellite operating in low Mars orbit. Mission-specific reference frames were derived for Sun-pointing, nadir-pointing, and GMO-pointing operations and integrated into an autonomous mission-management framework capable of selecting the appropriate spacecraft attitude mode according to orbital and mission conditions.

The spacecraft rotational dynamics, attitude-error evaluation algorithms, and proportional-derivative control architecture were implemented in MATLAB and tested through a series of increasingly complex simulations. Individual mode simulations demonstrated successful stabilization and tracking performance for both inertially fixed and rotating reference frames. The selected controller gains, designed to satisfy a 120-second closed-loop decay time constant, produced convergence behavior consistent with analytical expectations and provided effective damping of attitude and angular-velocity tracking errors.

The complete mission simulation demonstrated that the spacecraft can autonomously transition between operational modes while maintaining stable closed-loop performance throughout the mission timeline. Although temporary tracking errors were introduced following reference-frame changes, these transients remained bounded and decayed rapidly without compromising spacecraft stability or mission objectives. The results confirmed successful execution of Sun-pointing, nadir-pointing, and GMO-pointing operations within a unified control framework.

Independent validation was performed through reconstruction of the mission scenario in STK. The visual behavior observed during all operational phases showed strong agreement with the MATLAB simulations and verified the correctness of the implemented reference-frame definitions, attitude histories, and mission logic. The combined analytical, numerical, and visual results provide confidence in the overall spacecraft attitude-control architecture.

Future work may incorporate environmental disturbance torques, actuator and sensor limitations, attitude-determination algorithms, and advanced nonlinear control techniques to further improve performance and increase simulation fidelity. Nevertheless, the present work demonstrates a complete and successful implementation of autonomous attitude control for a multi-mode Mars-orbiting nano-satellite mission.

References

- [1] Hanspeter Schaub and John Junkins. *Analytical Mechanics of Space Systems*. American Institute of Aeronautics and Astronautics, Inc., 2018. ISBN: 9781624105210. DOI: <https://doi.org/10.2514/4.105210>.